Population Data Analysis SQL Project

This SQL project analyzes global population data to provide insights into growth trends, country-specific statistics, and regional comparisons. The dataset has been cleaned and structured to facilitate complex queries.

1. Data Preparation and Cleaning

Before analysis, the following steps were performed to ensure data integrity:

```
USE populection;

-- Clean column names
ALTER TABLE GLOBLE RENAME COLUMN entity TO COUNTRY;
ALTER TABLE GLOBLE RENAME COLUMN POPULETION_NO TO POPULATION_NO;

-- Categorize data records for better filtering
ALTER TABLE GLOBLE ADD COLUMN REC_TYPE VARCHAR(50);
SET SQL_SAFE_UPDATES = 0;

UPDATE GLOBLE SET REC_TYPE = 'CONTINENT' WHERE COUNTRY LIKE '%(UN)%';

UPDATE GLOBLE SET REC_TYPE = 'Category' WHERE COUNTRY IN (
    'High-income countries', 'World', 'Low-income countries',
    'More developed regions', 'Small island developing states (SIDS)',
    'Upper-middle-income countries', 'Lower-middle-income countries'
);

UPDATE GLOBLE SET REC_TYPE = 'COUNTRY' WHERE REC_TYPE IS NULL;
```

2. Analytical Queries

Analysis of Global Population Growth (2020-2021)

The following query calculates the percentage growth of population between 2020 and 2021 for all countries.

```
SELECT
    t1.COUNTRY,
    t1.POPULATION_NO AS Population_2020,
    t2.POPULATION_NO AS Population_2021,
    ROUND(((t2.POPULATION_NO - t1.POPULATION_NO) / t1.POPULATION_NO) * 100,
2) AS Percentage_Growth
FROM GLOBLE t1
JOIN GLOBLE t2 ON t1.COUNTRY = t2.COUNTRY
WHERE t1.year = 2020 AND t2.year = 2021 AND t1.REC_TYPE = 'COUNTRY'
ORDER BY Percentage_Growth DESC;
```

Top 10 Countries by Average Population (2013-2023)

```
SELECT
    COUNTRY,
    AVG(POPULATION_NO) AS Average_Population
FROM GLOBLE
WHERE year BETWEEN 2013 AND 2023 AND REC_TYPE = 'COUNTRY'
GROUP BY COUNTRY
ORDER BY Average_Population DESC
LIMIT 10;
```

Continent-wise Annual Population Change

Using window functions to analyze the year-on-year growth percentage for continents.

```
SELECT
    COUNTRY AS Continent,
    year,
    POPULATION_NO,
    LAG(POPULATION_NO) OVER (PARTITION BY COUNTRY ORDER BY year) AS
Previous_Year_Pop,
    ROUND(((POPULATION_NO - LAG(POPULATION_NO) OVER (PARTITION BY
COUNTRY ORDER BY year)) /
    LAG(POPULATION_NO) OVER (PARTITION BY COUNTRY ORDER BY year)) * 100, 2)
AS Growth_Percentage
```

```
FROM GLOBLE
WHERE REC_TYPE = 'CONTINENT'
ORDER BY Continent, year;
```